

Proposal: Solar + Battery Microgrid for Ecobank — ELF Branch

Client: Ecobank (client_id: 29)

Site: Eco Bank ELF Branch (branch_id: 87)

Objective: Cut diesel + grid spend, improve uptime, and stabilize power quality by placing the **whole branch load behind a 3-phase hybrid PCS/inverter** blending **Solar + Battery + Grid + Generator**.

1) Executive Summary (Compelling Offer)

Ecobank ELF is an excellent candidate for solar because:

- The branch has **strong daytime energy use** (very low risk of PV wastage).
- The site runs **a lot of generator**, even during daytime—so each solar kWh offsets **high-cost diesel**.

Recommendation (best ROI): deploy **~80 kWp PV + 100 kW hybrid PCS** now (fastest payback).

Upgrade path: add **100 kWh battery** when you want resilience / night diesel reduction.

2) What the Meter Data Says (Last 12 Months)

Analysis window: trailing 12 months (anchored to latest reading on **2026-01-21**), Lagos time.

2.1 Energy (kWh)

- **Average site energy:** ~519 kWh/day
 - **Daytime (09:00–16:30):** ~266 kWh/day
 - **Night/other hours:** ~253 kWh/day

Totals over period: 174,463 kWh

- **Generator:** 88,904 kWh (~51%)

- **Utility/grid: 85,559 kWh (~49%)**

2.2 Demand (kW)

For reliable design, we used percentiles (not raw spikes):

- Utility meter: **p99 $\approx$ 83 kW**, max observed $\approx$ 96 kW

Hybrid PCS sizing: 100 kW, 3-phase (covers observed peaks with margin).

3) System Architecture (Whole-Load Through the PCS)

Single protected bus: all branch loads are fed from the hybrid PCS output.

Operating logic:

1. Solar serves load first
2. Excess solar charges battery (if present)
3. PCS imports from grid when needed
4. During grid outages, PCS accepts/starts generator and blends seamlessly

Must-have electrical scope: maintenance bypass, protection coordination, generator interface (anti-backfeed / reverse power), monitoring + metering.

4) Recommended Sizing Options

Option A — PV-only (best ROI / fastest payback)

- **PCS:** 100 kW hybrid PCS/inverter
- **PV:** $\sim$ 80 kWp ($\approx$ 130 $\times$ 620W panels)

Option C — PV + 100 kWh battery (resilience + night diesel reduction)

- Same PV/PCS as Option A
- **Battery:** 100 kWh nominal

- Estimated battery night delivery from PV surplus: **~30 kWh/day**

Option B — Deeper offset (higher savings, slower payback)

- **PV: ~100 kWp** ($\approx 162 \times 620\text{W}$ panels)
- **Battery: ~200 kWh nominal**
- Target: ~70% energy offset (energy-basis)

5) Financial Model (ROI / Payback)

5.1 Energy value used (measured blend)

Based on measured generator vs utility shares:

- **Day avoided-cost: ~~~₦~~282/kWh**
- **Night avoided-cost: ~~~₦~~258/kWh**

5.2 PV yield basis

- **PSH = 5.0, PR = 0.75**
- Expected yield $\approx 1,369 \text{ kWh/kWp/year}$

6) CAPEX Breakdown (Benchmarks Provided; BOS/Installation Excluded)

Unit costs used: PCS **₦15.0m**; panels **₦150k/panel (620W)**; battery **₦19.2m per 60 kWh** ($\approx \text{₦320k/kWh}$).

6.1 Option A CAPEX (PV-only)

- PCS (100 kW): **₦15,000,000**
- Panels: **$130 \times \text{₦150,000} = \text{₦19,500,000}$**
- **Total CAPEX: ₦34,500,000**

6.2 Option C CAPEX (PV + 100 kWh)

- PCS (100 kW): ₦15,000,000
- Panels: ₦19,500,000
- Battery: $100 \text{ kWh} \times \text{₦}320,000/\text{kWh} = \text{₦}32,000,000$
- **Total CAPEX: ₦66,500,000**

6.3 Option B CAPEX (PV + 200 kWh)

- PCS (100 kW): ₦15,000,000
- Panels: $162 \times \text{₦}150,000 = \text{₦}24,300,000$
- Battery: $200 \text{ kWh} \times \text{₦}320,000/\text{kWh} = \text{₦}64,000,000$
- **Total CAPEX: ₦103,300,000**

7) Savings & Payback Summary

Option	PV	Battery	Annual savings (₦/yr)	CAPEX (₦)	Simple payback
A (Recommended)	~80 kWp	—	₦27.39m	₦34.50m	1.26 yrs
C	~80 kWp	100 kWh	₦30.25m	₦66.50m	2.20 yrs
B	~100 kWp	200 kWh	₦36.54m	₦103.30m	2.83 yrs

Compelling conclusion: Option A delivers the fastest payback because daytime demand is high and generator usage is material—solar offsets diesel immediately.

8) What's Excluded (to be priced after site survey)

- Mounting/BOS, cabling, combiner boxes, breakers, earthing/lightning
- Switchgear, ATS/generator interface, protection coordination

- Installation labor, civil works, permits, commissioning
- Optional remote monitoring / SLA

9) Next Steps (Fast Execution)

1. **Site survey (1 day):** roof layout, shading, structural checks, cable routes, generator interface
2. **Final engineering:** single-line, protection, BOS + installation “all-in” CAPEX
3. **Deployment:** procurement + installation + commissioning

Approval request: confirm whether you want **Option A now** with **Option C as the upgrade path**, or proceed directly to a battery-backed option for resilience.